UNITED STATES
NUCLEAR REGULATORY COMMISSION
WASHINGTON, D.C. 20555-0001

April 13, 2020

Mr. Bill Ross, Quality Assurance Manager
EnerSys
2366 Bernville Rd.
Reading, PA 19605

SUBJECT: NUCLEAR REGULATORY COMMISSION VENDOR INSPECTION REPORT OF ENERSYS, NO. 99901435/2020-201

Dear Mr. Ross:

From February 24 through February 28, 2020, the U.S. Nuclear Regulatory Commission (NRC) staff conducted an inspection at EnerSys' facilities in Hays, KS. The purpose of this limited-scope inspection was to assess EnerSys' compliance with the provisions of Title 10 of the *Code of Federal Regulations* (10 CFR) Part 21, "Reporting of Defects and Noncompliance," and selected portions of Appendix B, "Quality Assurance Program Criteria for Nuclear Power Plants and Fuel Reprocessing Plants," to 10 CFR Part 50, "Domestic Licensing of Production and Utilization Facilities."

This technically-focused inspection specifically evaluated EnerSys' implementation of the quality activities associated with fabrication, assembly, and testing of safety-related batteries and components being supplied to the U.S. operating nuclear power plants. The enclosed report presents the results of the inspection. This NRC inspection report does not constitute NRC endorsement of EnerSys' overall quality assurance (QA) or 10 CFR Part 21 programs.

Based on the results of this inspection, the NRC inspection team found the implementation of your QA program met the applicable technical and requirements imposed on you by your customers or NRC licensees. No findings of significance were identified.

In accordance with 10 CFR 2.390, "Public Inspections, Exemptions, Requests for Withholding," of the NRC's "Rules of Practice," the NRC will make available electronically for public inspection a copy of this letter, its enclosure, and your response through the NRC Public Document Room or from the NRC's Agencywide Documents Access and Management System, which is accessible at http://www.nrc.gov/reading-rm/adams.html.

EXECUTIVE SUMMARY

ENERSYS
99901435/2020-201

The U.S. Nuclear Regulatory Commission (NRC) staff conducted a vendor inspection at the EnerSys (hereafter referred to as EnerSys) facility in Hays, KS to verify that it had implemented an adequate quality assurance (QA) program that complies with the requirements of Appendix B, "Quality Assurance Criteria for Nuclear Power Plants and Fuel Reprocessing Plants," to Title 10 of the *Code of Federal Regulations* (10 CFR) Part 50, "Domestic Licensing of Production and Utilization Facilities" and 10 CFR Part 21, "Reporting of Defects and Noncompliance." This was the second NRC vendor inspection at EnerSys (see NRC Inspection Report 99901435/2013-201 Agencywide Documents Access and Management System (ADAMS) No. ML14058A705).

This technically-focused inspection specifically evaluated EnerSys' implementation of the quality activities associated with the fabrication, assembly, and testing of safety-related batteries and components being suppling to U.S. operating nuclear power plants and the AP1000 reactor design.

These regulations served as the bases for the NRC inspection:

- Appendix B to 10 CFR Part 50
- 10 CFR Part 21

During the course of this inspection, the NRC inspection team implemented inspection procedure (IP) 43002, "Routine Inspections of Nuclear Vendors," dated January 27, 2017, IP 43003, "Reactive Inspections of Nuclear Vendors," dated December 14, 2015, IP 43004, "Inspection of Commercial-Grade Dedication Programs," dated January 27, 2017, IP 36100, "Inspection of 10 CFR Part 21 and Programs for Reporting of Defects and Noncompliance," dated February 13, 2012, and IP 71152, "Problem Identification and Resolution," Appendix 1, "Guidance for Gathering SCWE and PI&R Insights," dated February 26, 2015.

The NRC inspection team concluded that EnerSys' QA policies and procedures comply with the applicable requirements of Appendix B to 10 CFR Part 50 and 10 CFR Part 21, and that EnerSys' personnel are implementing these policies and procedures effectively. The results of this inspection are summarized below.

Nonconforming Materials, Parts, or Components and Corrective Action

The NRC inspection team reviewed EnerSys' policies and implementing procedures that govern the Nonconformance of materials, parts and components and Corrective Action programs. The NRC inspection team reviewed a sample of Material Review Reports (MRRs), supplier quality alerts and corrective actions reports (CARs). The NRC inspection team noted that evaluation of corrective actions related to nuclear grade batteries is performed by personnel at EnerSys Headquarters in Reading, PA. While the NRC inspection team did not identify recurring instances of untimely or inadequate evaluation of MRRs and/or CARs, the NRC inspection team did notice that EnerSys in Hays and Headquarters failed to open their own CARs to address the Supplier Corrective Actions Reports (SCARs) that Westinghouse documented in their external audit report for the audit performed at EnerSys in Hays. This oversight was considered a minor issue by the NRC inspection team. EnerSys opened six corrective action reports to address this

issue, CAR150252-022020, CAR150253-022020, CAR150254-022020, CAR150255-022020, CAR150256-022020, CAR150257-022020, CAR150258-022020. No findings of significance were identified.

Audits

The NRC inspection team reviewed EnerSys' policies and implementing procedures that govern the audit program to verify compliance with the requirements of Criterion XVIII, "Audits," of Appendix B to 10 CFR Part 50. Currently, EnerSys at Hays, KS does not perform any external audits as all their supplier are commercial suppliers. The NRC inspection team reviewed a sample of EnerSys internal audits. The NRC inspection team noted that the audits conducted at EnerSys at Hays, KS are performed by personnel from EnerSys Headquarters in Reading, PA and that the EnerSys personnel performing the audits are appropriately trained personnel not having direct responsibilities in the areas being audited. No findings of significance were identified

Design Control and Commercial-Grade Dedication

The NRC inspection team reviewed EnerSys' policies and implementing procedures that govern the commercial-grade dedication program to verify compliance with the requirements of Criterion III, "Design Control," of Appendix B to 10 CFR Part 50. The NRC inspection team reviewed a sample of EnerSys procurement documents and commercial-grade dedication packages. No findings of significance were identified.

Safety Conscious Work Environment (SCWE)

The NRC inspection team concluded that EnerSys' SCWE program and implementation were consistent with the NRC's guidance in IP 71152, "Problem Identification and Resolution," Appendix 1, "Guidance for Gathering SCWE and PI&R Insights." Based on interviews conducted of selected individuals within EnerSys' organization, the NRC inspection team determined that individuals were comfortable, and willing to raise safety concerns to their supervisors and senior management. No findings of significance were identified.

REPORT DETAILS

1. Nonconforming Materials, Parts, or Components and Corrective Action

   a. Inspection Scope

   The U.S. Nuclear Regulatory Commission (NRC) inspection team reviewed EnerSys' policies and implementing procedures that govern the control of nonconformances and corrective action to verify compliance with the requirements of Criterion XV, "Nonconforming Materials, Parts, or Components," and Criterion XVI, "Corrective Action," of Appendix B, "Quality Assurance Criteria for Nuclear Power Plants and Fuel Reprocessing Plants," to Title 10 of the *Code of Federal Regulations* (10 CFR) Part 50, "Domestic Licensing of Production and Utilization Facilities," respectively. The NRC inspection team reviewed EnerSys' Material Review Report (MRR) log and reviewed a sample of MRRs to ensure that EnerSys implemented an adequate program to assess and control nonconforming items, including appropriate identification, documentation, segregation, evaluation and disposition. Additionally, the NRC inspection team interviewed EnerSys personnel to verify there were designated areas to segregate and control nonconforming materials. The NRC inspection team also toured the facility and saw several areas where nonconforming material was segregated.

   At Enersys, Corrective Action Reports (CARs) are designated by the severity type and root cause determination is warranted for the CARs with higher severity type. The NRC inspection team discussed the MRRs and corrective action programs with Enersys' management and technical personnel. The attachment to this inspection report lists the documents reviewed and personnel interviewed by the NRC inspection team.

   b. Observations and Findings

   Westinghouse, as a customer of EnerSys at Hays, KS performed an external audit of EnerSys at both locations, Reading, PA and Hays, KS. During the audit at Hays, KS, Westinghouse identified five issues that were documented as Supplier Corrective Action Requests (SCARs) in the subsequent audit report. Westinghouse's due date for these SCARs was February 9, 2020 and three of the SCARs had a "No Product Shipment" note attached to them due to the significance of the issues identified. The NRC inspection team noted that EnerSys failed to document and open CARs for the issues at the Hays, KS facility to track and implement timely corrective actions to address these five SCARs. Specifically, while reviewing a sample of past internal audits performed at EnerSys at Hays, KS and another external audit performed at EnerSys at the Reading, PA location, the NRC inspection team verified that EnerSys opened their own CARs to correspond and address the issues found during these audits as described in their quality assurance (QA) procedures. However, EnerSys Hays, KS facility failed to open corresponding CARs for the SCARs identified during the December 2019 Westinghouse audit. Although the Westinghouse's due date for responding to these SCARs was February 9, 2020, EnerSys did not open, track, respond or ask for an extension of the due date, as described in EnerSys' procedures for the SCARs identified at the Hays, KS facility.

c. Conclusion

While the NRC inspection team did not identify recurring instances of untimely or inadequate evaluation of MRRs and/or CARs, the NRC inspection team did notice that an EnerSys facility failed to open CARs to address the SCARs that the purchaser documented in their external audit report for the audit performed at EnerSys in Hays, KS. This oversight was considered a minor issue by the NRC inspection team as there was no impact to nuclear work activities at the EnerSys Hays facility at the time of the inspection. EnerSys opened six CARs to address this issue; CAR150252-022020, CAR150253-022020, CAR150254-022020, CAR150255-022020, CAR150256-022020, CAR150257-022020, and CAR150258-022020. No findings of significance were identified.

2. Audits

a. Inspection Scope

The NRC inspection team reviewed EnerSys' policies and implementing procedures that govern its audit program to verify compliance with the requirements of Criterion XVIII, "Audits," and Criterion VII, "Control of Purchased Material, Equipment, and Services," of Appendix B to 10 CFR Part 50. The NRC inspection team reviewed a sample of internal audits to verify the scope, frequency and independence of the auditors from the topic they were auditing. Additionally, the NRC inspection team reviewed EnerSys' Approved Suppliers List (ASL) and selected a sample of suppliers to review the methodology of conducting and documenting commercial grade surveys (CGSs) to verify adequate evaluation of the sub-supplier's quality practices. The NRC inspection team reviewed EnerSys' process for conducting CGSs at an established frequency. The NRC inspection team verified that EnerSys had prepared and approved plans that identified the audit scope and applicable checklist criteria before the initiation of the audit activity. The NRC inspection team also discussed the audit program and performance of CGSs with EnerSys' management and technical staff. The attachment to this inspection report lists the documents reviewed and personnel interviewed by the NRC inspection team.

b. Observation and Findings

No findings of significance were identified.

c. Conclusion

The NRC inspection team concluded that EnerSys is implementing its audit program and CGSs in accordance with the regulatory requirements of Criterion XVIII and and Criterion VII, respectively of Appendix B to 10 CFR Part 50. Based on the limited sample of documents reviewed and activities observed, the NRC inspection team determined that EnerSys is implementing its policies and procedures associated with internal audit program and CGSs. No findings of significance were identified.

3. Design Control and Commercial-Grade Dedication

a. Inspection Scope

The NRC inspection team reviewed EnerSys's policies and implementing procedures that govern the implementation of its commercial-grade dedication (CGD) to verify compliance with the requirements of Criterion III, "Design Control," and Criterion VII of Appendix B to 10 CFR Part 50. The NRC inspection team also discussed the qualification and CGD with the Enersys' technical personnel located in Reading, PA and manufacturing technical personnel on-site.

Qualification

The NRC inspection team reviewed the design control of EnerSys' nuclear-qualified battery (GN product type), GN related purchased parts and auxiliary components. EnerSys GN parts use the "CERT" suffix designation for nuclear safety-related applications. EnerSys GN original qualification was conducted by an external laboratory. The nuclear environmental qualification test included GN-13, GN-15, GN-17 GN-23 and GN-29 cell types and simulated accelerated aging groups. The NRC inspection team also reviewed purchase orders for the GN-23 and GN-29 battery systems to ensure they included the appropriate regulatory requirements and product qualifications specified by the customers in accordance with applicable Class 1E batteries for nuclear power generating stations' Institute of Electrical and Electronics Engineers (IEEE) qualification standards.

Commercial Grade Dedication

The NRC inspection team reviewed the CGD methodology for items and services to be installed in Class 1E batteries, including the development of critical characteristics (CCs), technical evaluations, failure mode and effects analysis, acceptance criteria methods, sampling methodology, checklists, reports, and associated Purchase Orders. The NRC inspection team reviewed the CGD process for the GN-23 and GN-29 product types. The NRC inspection team evaluated a sample of technical evaluations and concluded that the technical evaluations in the dedication methodology appropriately identified the CCs necessary to provide reasonable assurance that the item or service would perform its intended safety function.

The NRC inspection team reviewed a sample of commercial-grade dedicated battery accessories and components supplied as basic components and controlled under EnerSys' Appendix B program. Accessories included, but were not limited to rack assemblies, bolt packages, terminal plates, lugs, washers, and cables. The NRC inspection team reviewed EnerSys sampling methodology and the GN product type single sampling plan for inspections. The attachment to this inspection report lists the documents reviewed and personnel interviewed by the NRC inspection team.

b. Observation and Findings

No findings of significance were identified

c. Conclusion

The NRC inspection team concluded that EnerSys is implementing its CGD program activities in accordance with the regulatory requirements of Criterion III and Criterion VII of Appendix B to 10 CFR Part 50. Based on the limited sample of documents reviewed and activities observed, the NRC inspection team determined that EnerSys is

implementing its policies and procedures associated with CGD program and oversight of contracted activities. No findings of significance were identified.

4. Safety Conscious Work Environment (SCWE)

a. Inspection Scope

The NRC inspection team determined that the EnerSys personnel are willing to raise nuclear safety concerns and have not observed any hesitation of their coworkers in raising safety concerns. EnerSys personnel also indicated that they felt comfortable raising concerns to their supervisor and management, and elevating issues up through supervision or management if not appropriately addressed. Most of Enersys' personnel prefer to submit issues through their supervisor.

The NRC inspection team and EnerSys management discussed EnerSys' safety culture and SCWE program. EnerSys' personnel work to a high level of skill-of-the-craft and the interviewed personnel have comfortable and approachable relationships with management which promotes an adequate SCWE. The NRC inspection team recommended that the EnerSys, Hays KS facility strengthen the safety culture by clarifying the SCWE program to better communicate EnerSys expectations for personnel to participate in the program.

b. Observations and Findings

No findings of significance were identified.

c. Conclusion

The NRC inspection team concluded from their observations that EnerSys safety culture was adequate.

5. Entrance and Exit Meetings

On February 24, 2020, the NRC inspection team discussed the scope of the inspection with Mr. Bill Ross, QA Manager, and other members of EnerSys' management and technical personnel. On February 28, 2020, the NRC inspection team presented the inspection results and observations during an exit meeting with Mr. Bill Ross, and other members of EnerSys' management and technical personnel. The attachment to this report lists the attendees of the entrance and exit meetings, as well as those individuals whom the NRC inspection team interviewed.